App Engine, Services, APIs, Github

PAAS

Quotas

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https://cloud.google.com/appengine/docs/quotas?hl=en_US
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Instance hours

Instance usage is billed by instance uptime, at a given hourly rate. Billable time starts when an instance starts, and ends fifteen minutes after it shuts down. There is no billing for idle instances above the maximum number of idle instances set in the Performance Settings tab of the Admin Console.

There are separate free daily quotas for frontend and backend instances. Note that when you use the Modules API, the module's instance class determines which quota applies.

Resource or API Call	Free Quota
Frontend Instances (Automatic Scaling Modules)	28 free instance-hours per day
Backend Instances (Basic and Manual Scaling Modules)	8 free instance-hours per day

Logs

The Logs API is metered when log data is retrieved, and is available for both paid and free apps.

Logs storage contains request logs and application logs for an application, and is available for both paid and free apps. For paid apps, you can increase total logs storage size and/or log data retention time, using the Log Retention setting in the **Admin Console**.

Resource	Free Default Limit	Billing Enabled Default Limit
Logs data retrieval	100 megabytes	No maximum for paid app.
Logs data	1 gigabyte	Log data kept for a maximum of 365 days if paid, 90 days if free.

Allows Growth

- Datastore
- GCS
- Compute Engine
- Managed VM’s (IAAS++)
- Cloud SQL
- memcache

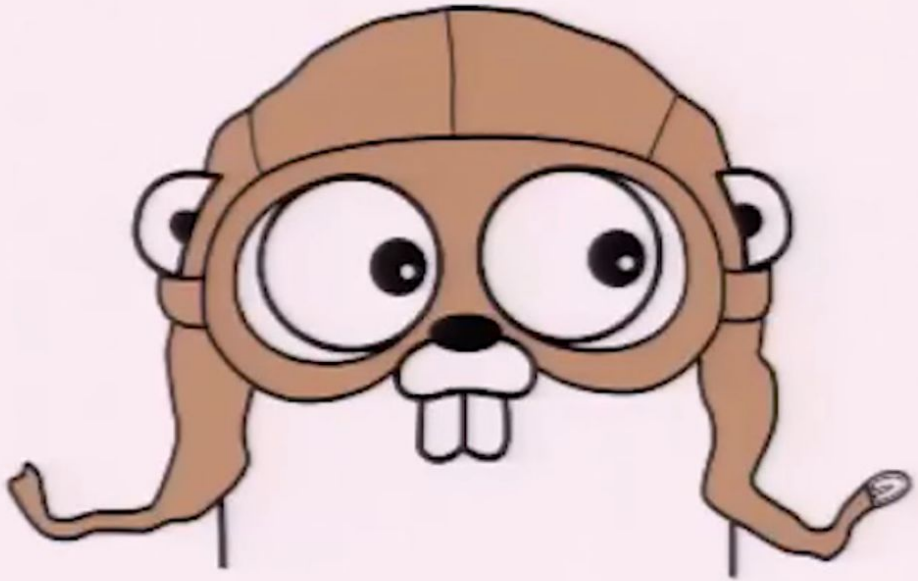
sandbox environment

Sandbox

- Can’t use the “unsafe” package
- can’t write files
  - use cloud storage instead
- outbound http requests
  - use the services they provide to make outbound http requests

Go + App Engine

- Go is awesome
- App Engine is an auto-scaling low-maintenance Platform-as-a-Service (give us your app's code, we'll run it on Google's servers), provides a bunch of services
- Most efficient runtime on App Engine: starts fast, runs fast, uses minimal RAM



Go + App Engine

- Go was open sourced in November 2009
- Go + App Engine development started in January 2011
- Go 1 arrived in March 2012
- Stable releases did not exist (`release.r56` was later)
- Makefiles and direct `6g` and `6l` invocation was the standard
- The `go` tool didn't exist, and package layout was not understood

Interesting differences to other runtimes

- Two runtimes before Go: Python (2.5) and Java
- Several interesting things we did differently for Go

Difference 1: Explicit Context

API calls (e.g. memcache, mail, logging) require a "security ticket"

- A security ticket is an opaque string, unique for every running request
- Used for quotas, some auth restrictions, associating logging
- Python has no concurrency, so security ticket is in global state
- Java uses thread-local storage
- Go is concurrent and avoids TLS -> explicit `Context`

`appengine.Context` is an interface; different concrete types in prod and dev

Difference 1: Explicit Context

An explicit `Context` interface had some unanticipated benefits:

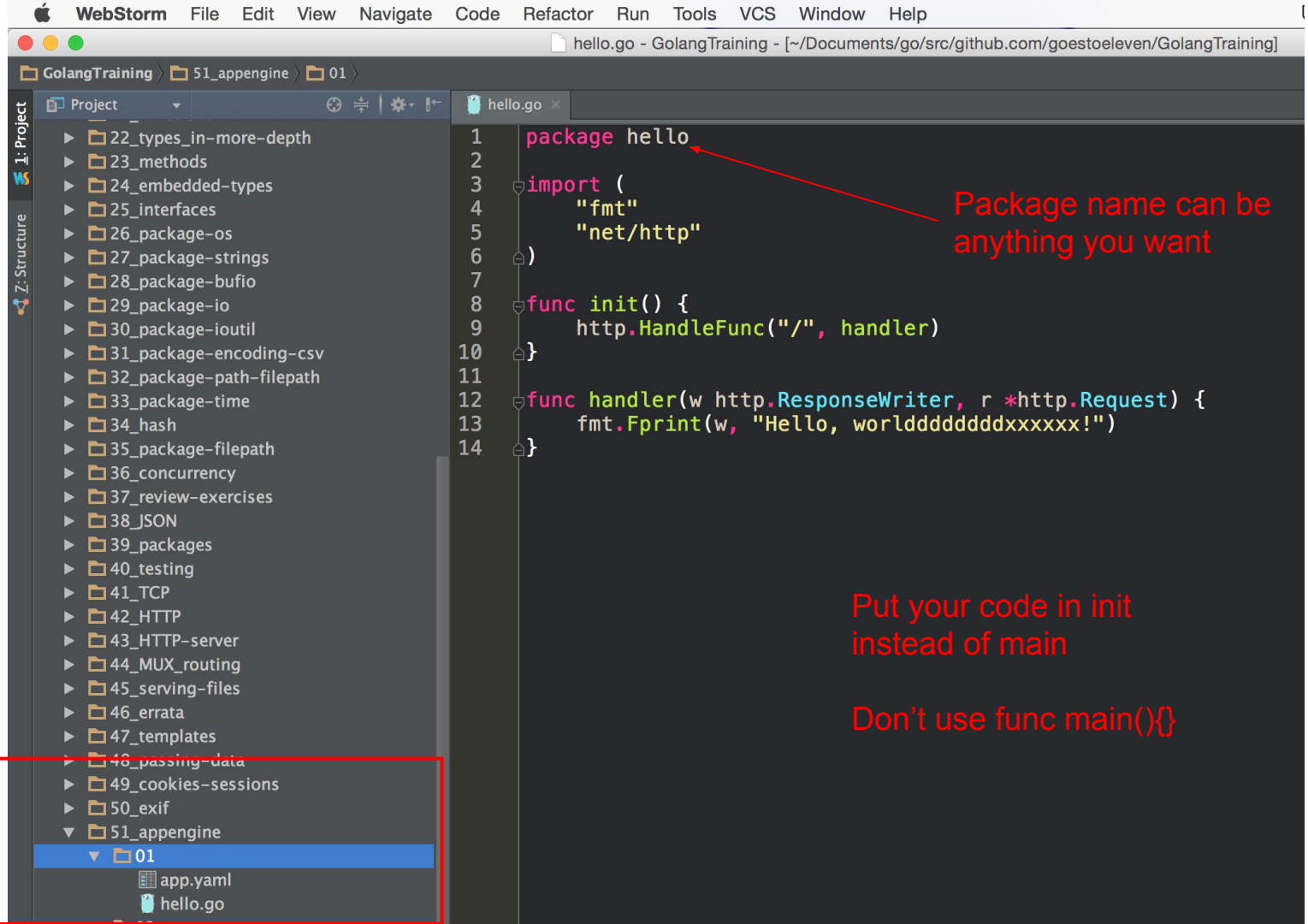
- Overriding default timeouts was easy: a function returning a new Context that wraps a provided Context
- Namespace support ("multitenancy": partitioned datastore, etc.) was the same
- Transaction support was the same
- ... all without having to change N APIs to support these features
- Interface design supported *orthogonality*

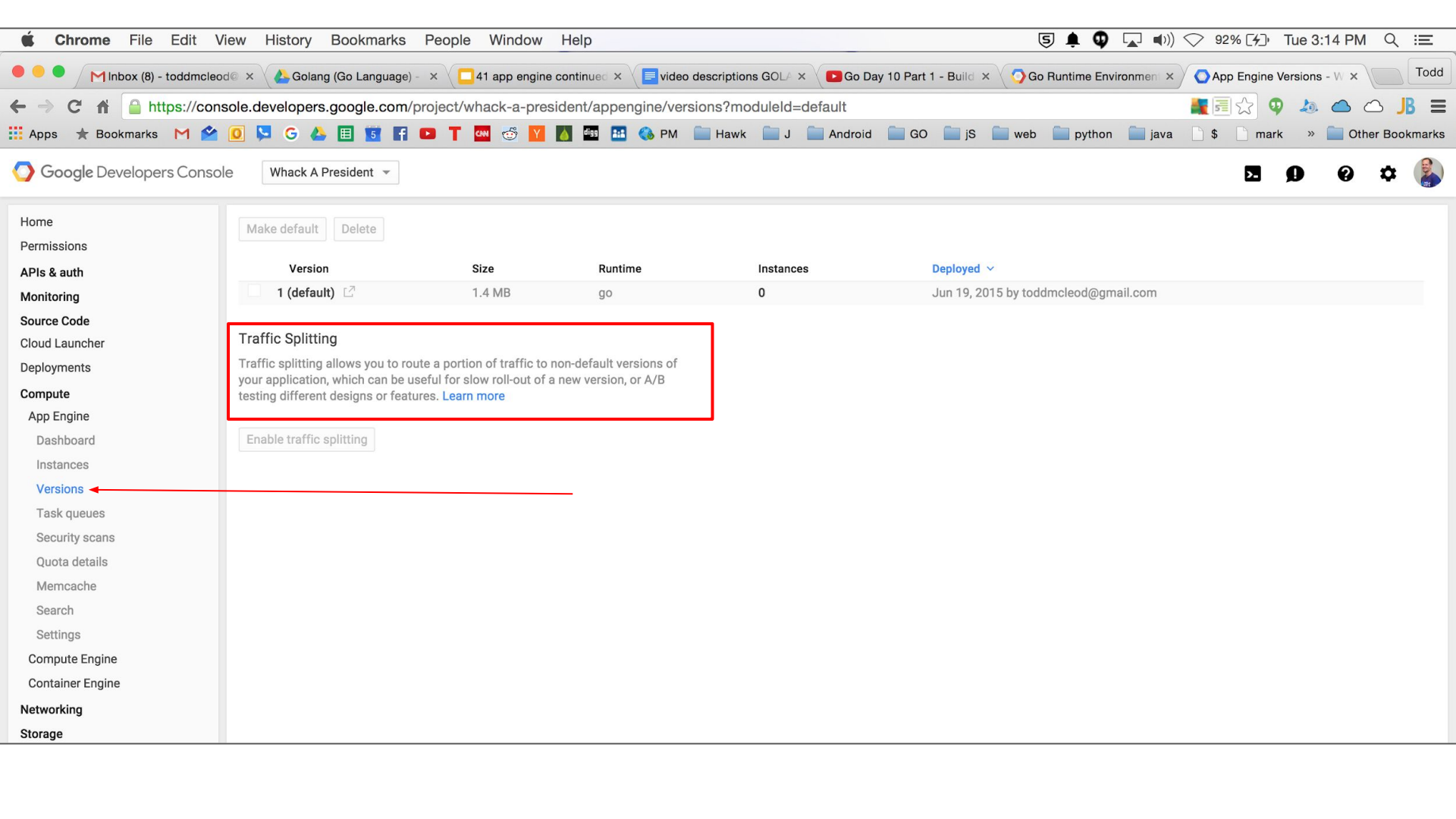
Go on App Engine

- Much easier than other languages
  - Python
  - Java
  - Php
- Go
  - Just the net/http package with go

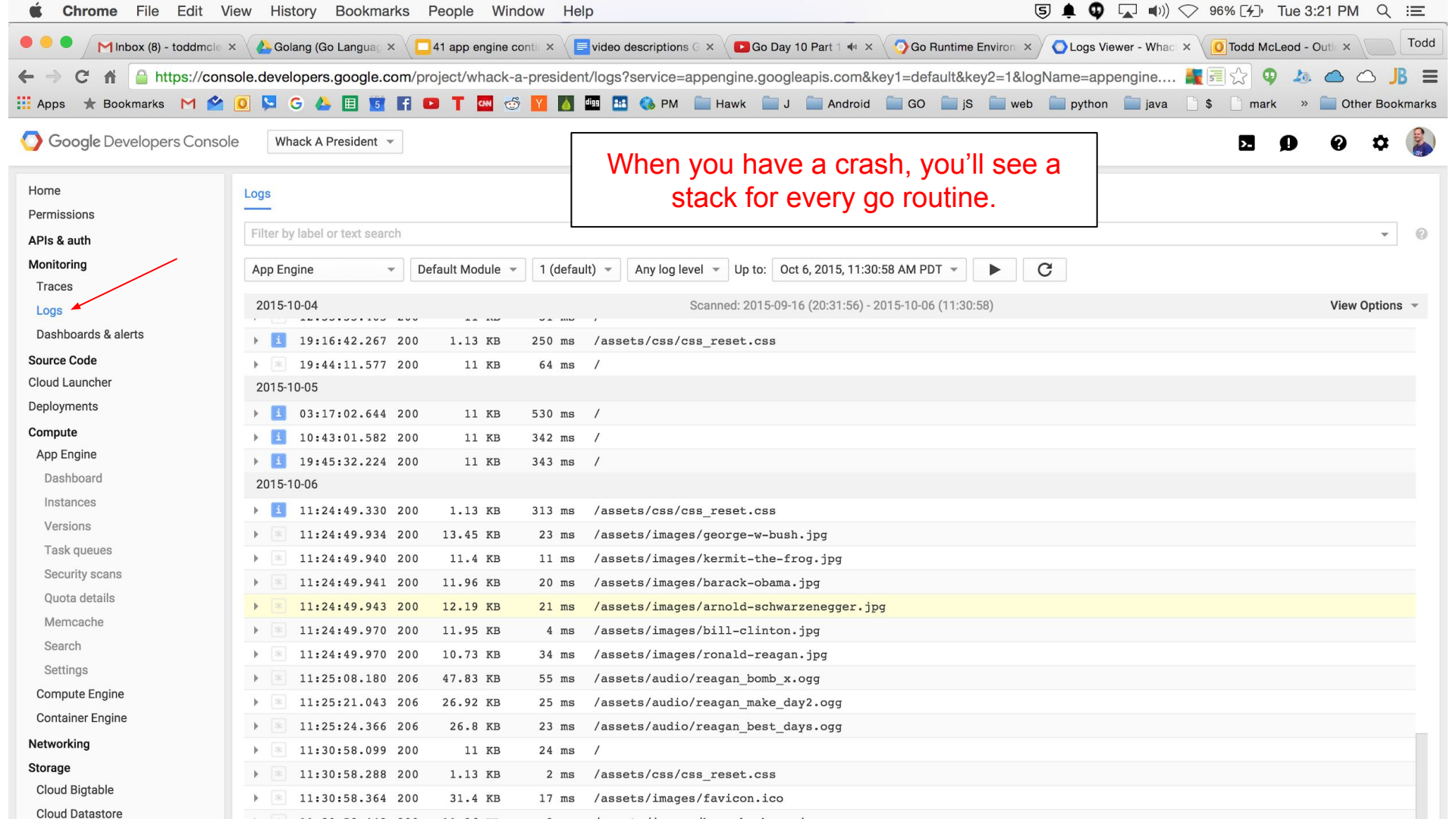
vendor lock-in

app layout


WebStorm File Edit View Navigate Code Refactor Run Tools VCS Window Help
hello.go - GolangTraining - [~/Documents/go/src/github.com/goestoeleven/GolangTraining]
GolangTraining 51_appengine 01
Project
hello.go
22_types_in-more-depth
23_methods
24_embedded-types
25_interfaces
26_package-os
27_package-strings
28_package-bufio
29_package-io
30_package-ioutil
31_package-encoding-csv
32_package-path-filepath
33_package-time
34_hash
35_package-filepath
36_concurrency
37_review-exercises
38_JSON
39_packages
40_testing
41_TCP
42_HTTP
43_HTTP-server
44_MUX_routing
45_serving-files
46_errata
47_templates
49_cookies-sessions
50_exif
51_appengine
01
app.yaml
hello.go
1: Project
Z: Structure
1 package hello
2
3 import (
4 "fmt"
5 "net/http"
6)
7
8 func init() {
9 http.HandleFunc("/", handler)
10 }
11
12 func handler(w http.ResponseWriter, r *http.Request) {
13 fmt.Fprint(w, "Hello, worlddddddddxxxxxx!")
14 }
Package name can be anything you want
Put your code in init instead of main
Don’t use func main(){}


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Go Runtime Environmen
App Engine Versions - W
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https://console.developers.google.com/project/whack-a-president/appengine/versions?moduleId=default
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python
java
$
mark
Other Bookmarks
Google Developers Console
Whack A President
Home
Permissions
APIs & auth
Monitoring
Source Code
Cloud Launcher
Deployments
Compute
App Engine
Dashboard
Instances
Versions
Task queues
Security scans
Quota details
Memcache
Search
Settings
Compute Engine
Container Engine
Networking
Storage
Make default
Delete
Version
Size
Runtime
Instances
Deployed
1 (default)
1.4 MB
go
0
Jun 19, 2015 by toddmcleod@gmail.com
Traffic Splitting
Traffic splitting allows you to route a portion of traffic to non-default versions of your application, which can be useful for slow roll-out of a new version, or A/B testing different designs or features. Learn more
Enable traffic splitting

logs


When you have a crash, you'll see a stack for every go routine.
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APIs & auth
Monitoring
Traces
Logs
Dashboards & alerts
Source Code
Cloud Launcher
Deployments
Compute
App Engine
Dashboard
Instances
Versions
Task queues
Security scans
Quota details
Memcache
Search
Settings
Compute Engine
Container Engine
Networking
Storage
Cloud Bigtable
Cloud Datastore
Logs
Filter by label or text search
App Engine
Default Module
1 (default)
Any log level
Up to: Oct 6, 2015, 11:30:58 AM PDT
2015-10-04
Scanned: 2015-09-16 (20:31:56) - 2015-10-06 (11:30:58)
View Options
19:16:42.267 200 1.13 KB 250 ms /assets/css/css_reset.css
19:44:11.577 200 11 KB 64 ms /
2015-10-05
03:17:02.644 200 11 KB 530 ms /
10:43:01.582 200 11 KB 342 ms /
19:45:32.224 200 11 KB 343 ms /
2015-10-06
11:24:49.330 200 1.13 KB 313 ms /assets/css/css_reset.css
11:24:49.934 200 13.45 KB 23 ms /assets/images/george-w-bush.jpg
11:24:49.940 200 11.4 KB 11 ms /assets/images/kermit-the-frog.jpg
11:24:49.941 200 11.96 KB 20 ms /assets/images/barack-obama.jpg
11:24:49.943 200 12.19 KB 21 ms /assets/images/arnold-schwarzenegger.jpg
11:24:49.970 200 11.95 KB 4 ms /assets/images/bill-clinton.jpg
11:24:49.970 200 10.73 KB 34 ms /assets/images/ronald-reagan.jpg
11:25:08.180 206 47.83 KB 55 ms /assets/audio/reagan_bomb_x.ogg
11:25:21.043 206 26.92 KB 25 ms /assets/audio/reagan_make_day2.ogg
11:25:24.366 206 26.8 KB 23 ms /assets/audio/reagan_best_days.ogg
11:30:58.099 200 11 KB 24 ms /
11:30:58.288 200 1.13 KB 2 ms /assets/css/css_reset.css
11:30:58.364 200 31.4 KB 17 ms /assets/images/favicon.ico

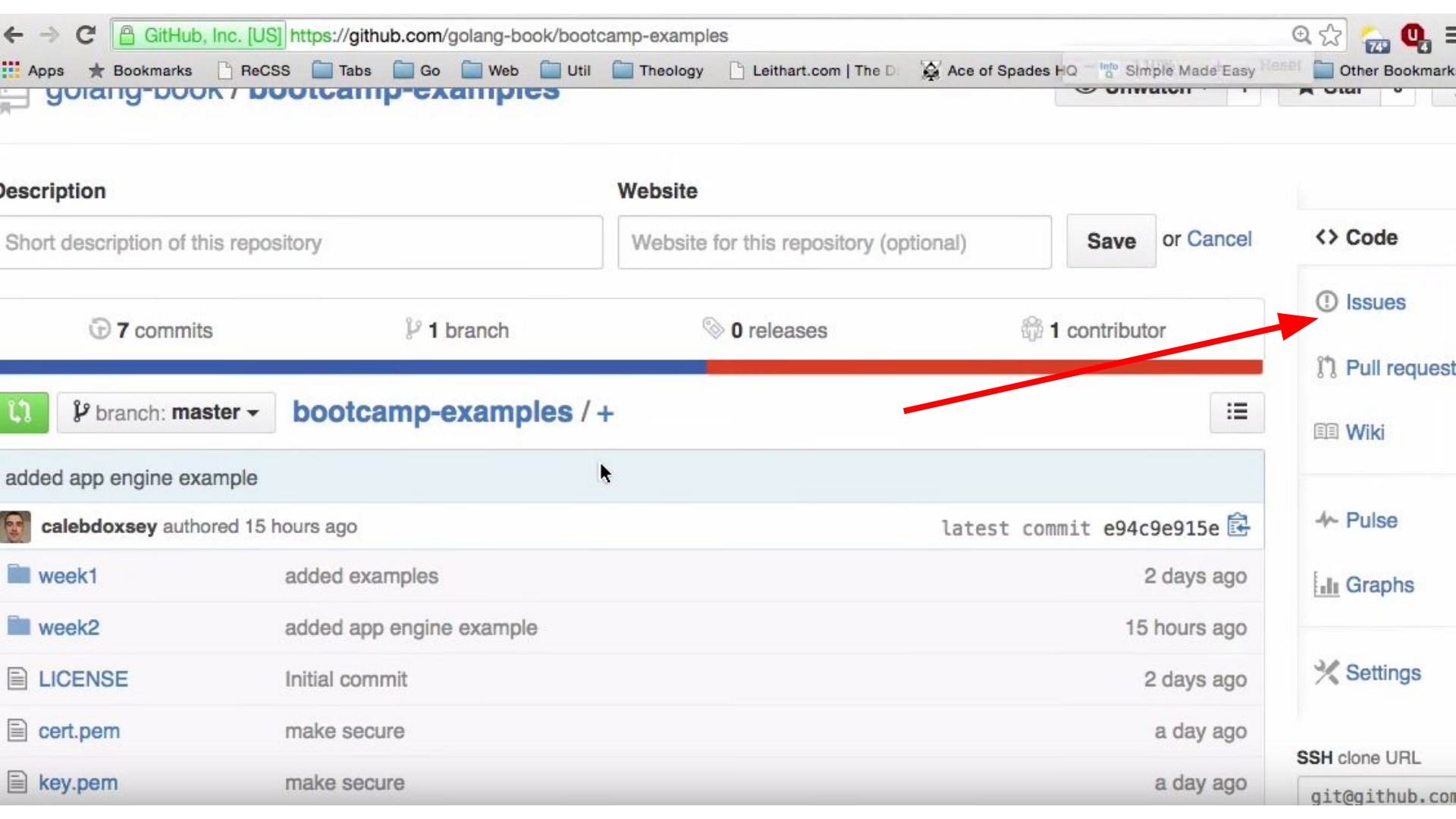
deploy

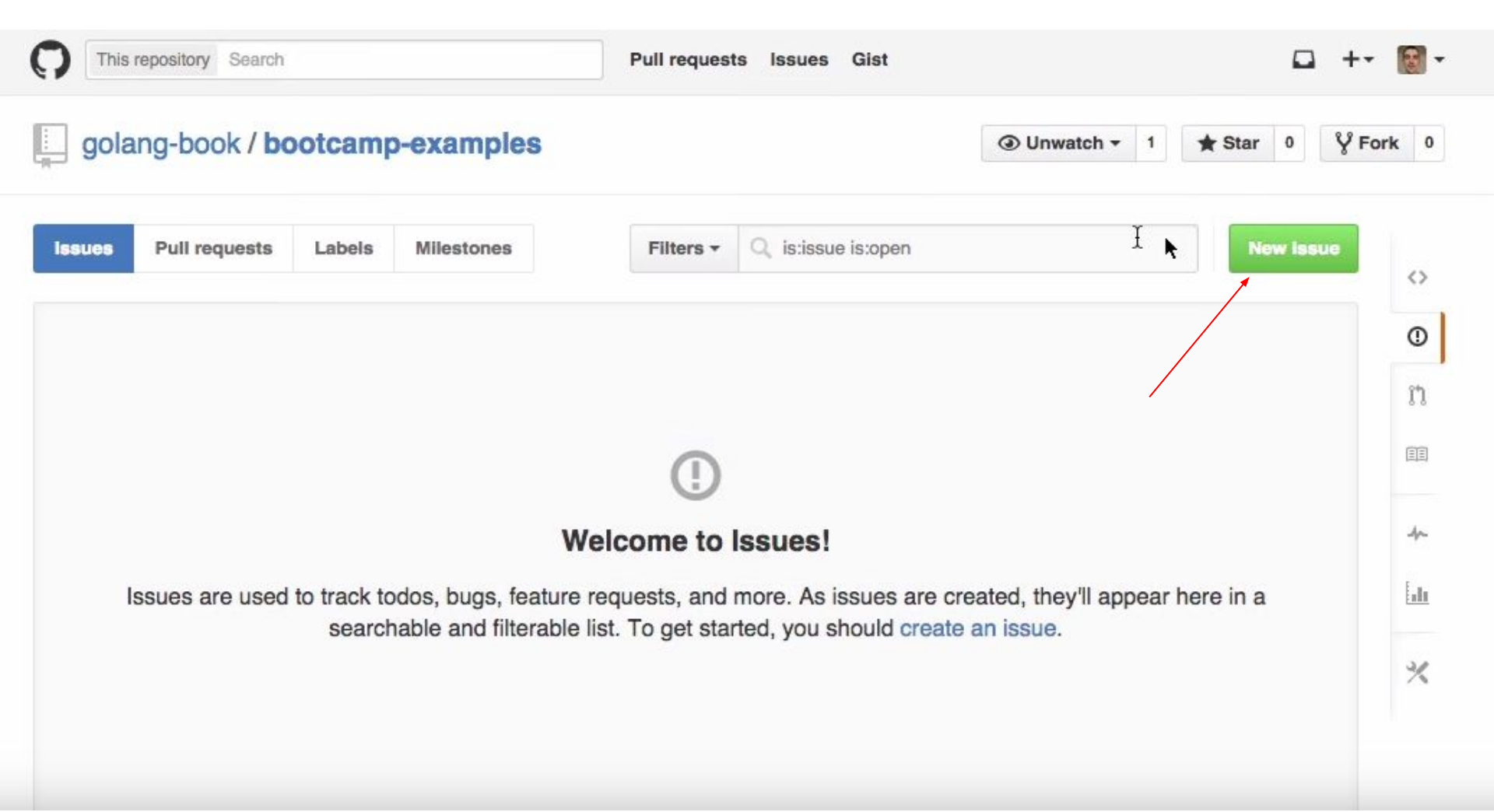
- instead of `goapp deploy`
  - appcfg.py update --oauth2 ./

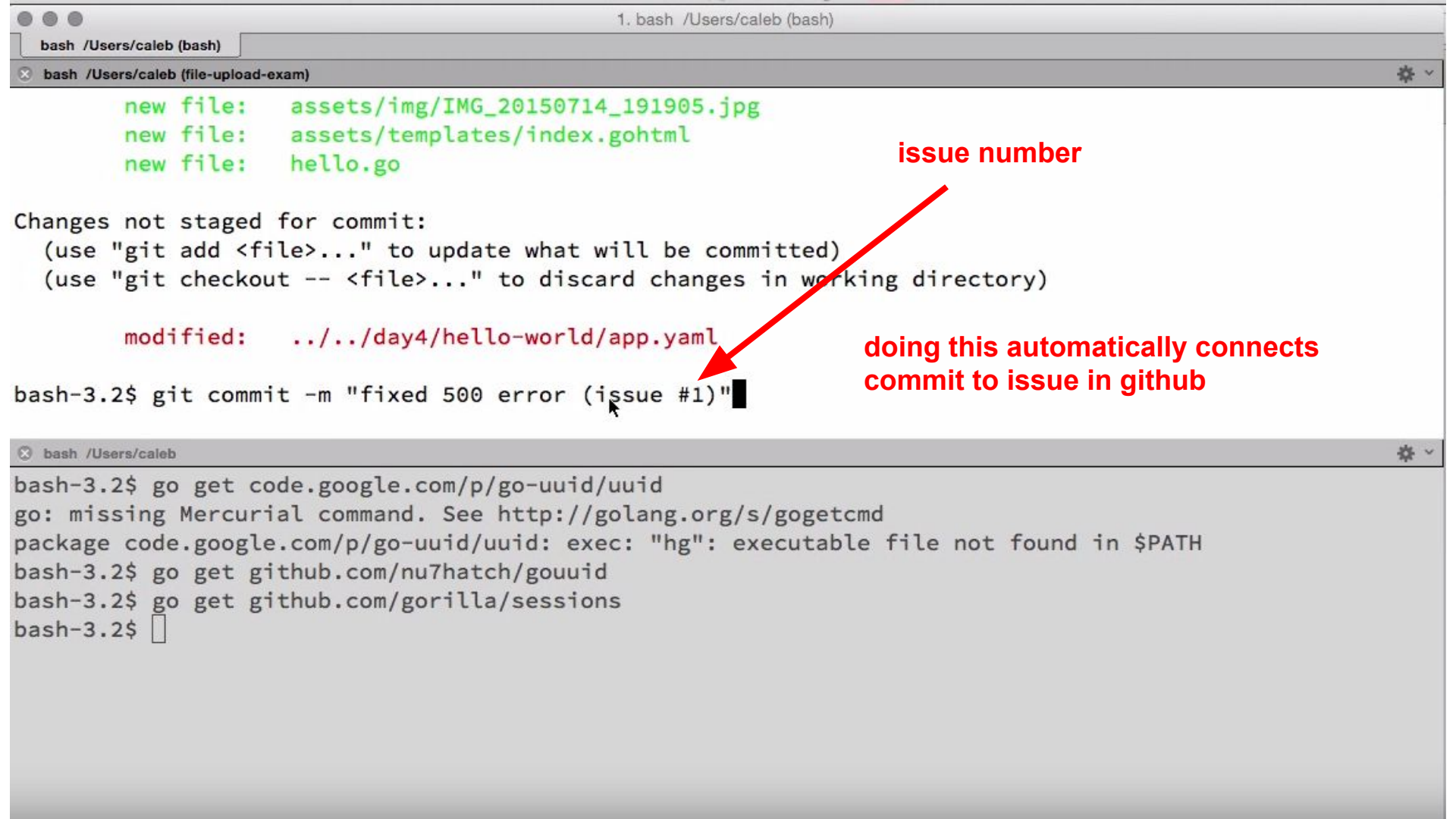
github

Typical process

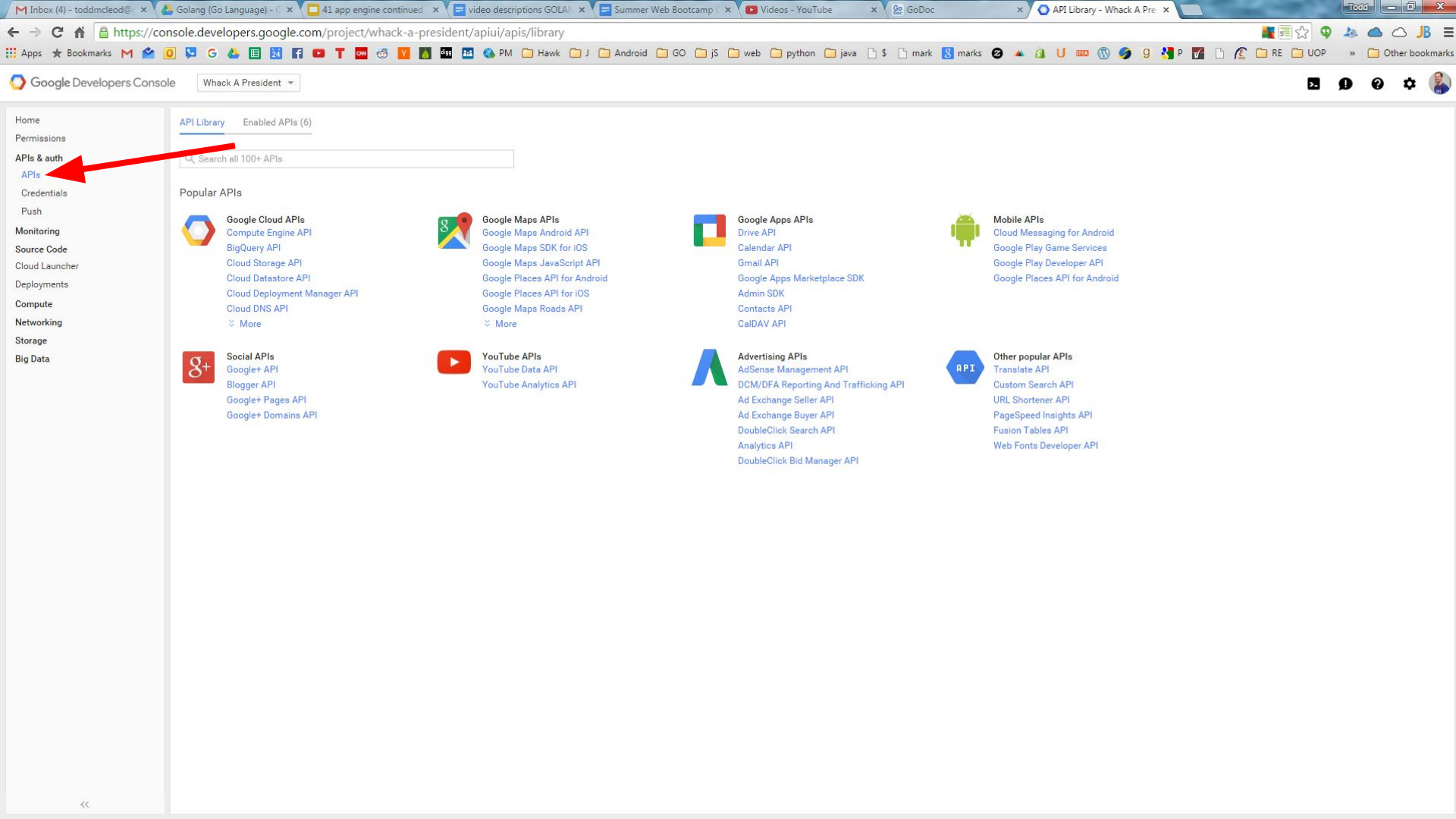
- create a github branch
- make a change
- commit it
- push it
- code review
- merge it to master
- redeploy


GitHub, Inc. [US] https://github.com/golang-book/bootcamp-examples
Apps Bookmarks ReCSS Tabs Go Web Util Theology Leithart.com | The D Ace of Spades HQ Simple Made Easy Other Bookmark
golang-book / bootcamp-examples
Description
Short description of this repository
Website
Website for this repository (optional)
Save or Cancel
7 commits 1 branch 0 releases 1 contributor
branch: master
bootcamp-examples / +
added app engine example
calebdoxsey authored 15 hours ago
latest commit e94c9e915e
week1 added examples 2 days ago
week2 added app engine example 15 hours ago
LICENSE Initial commit 2 days ago
cert.pem make secure a day ago
key.pem make secure a day ago
Code
Issues
Pull request
Wiki
Pulse
Graphs
Settings
SSH clone URL
git@github.com

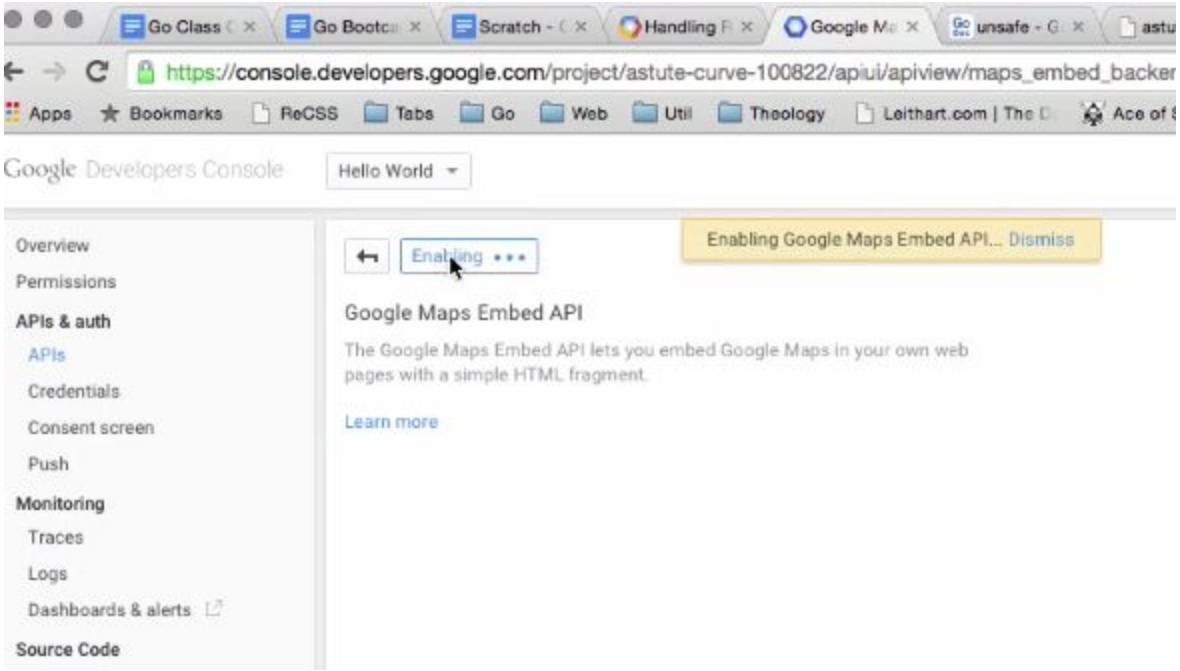

This repository
Search
Pull requests
Issues
Gist
golang-book / bootcamp-examples
Unwatch 1
Star 0
Fork 0
Issues
Pull requests
Labels
Milestones
Filters
is:issue is:open
New issue
Welcome to Issues!
Issues are used to track todos, bugs, feature requests, and more. As issues are created, they'll appear here in a searchable and filterable list. To get started, you should create an issue.

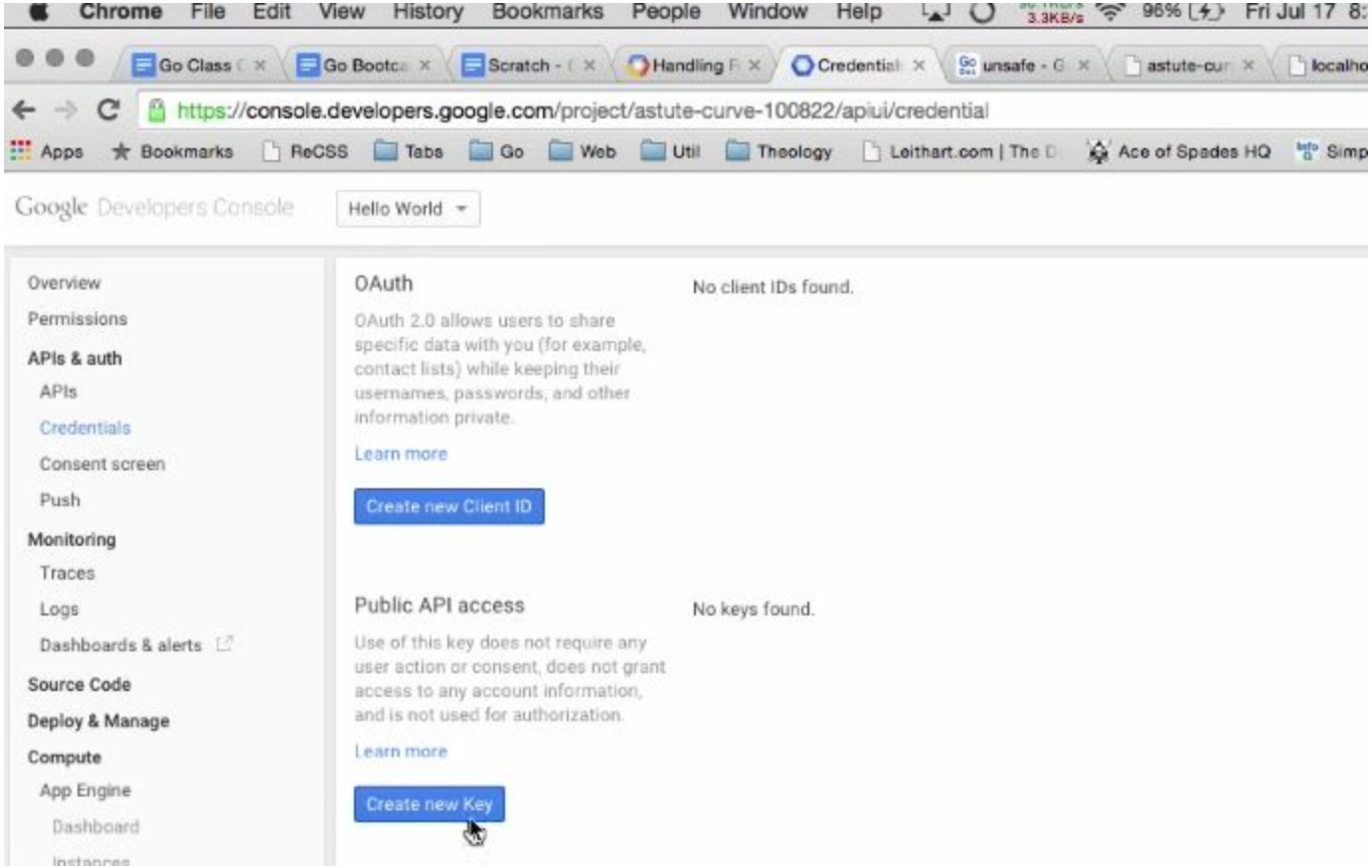

1. bash /Users/caleb (bash)
bash /Users/caleb (bash)
bash /Users/caleb (file-upload-exam)
new file: assets/img/IMG_20150714_191905.jpg
new file: assets/templates/index.gohtml
new file: hello.go
Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git checkout -- <file>..." to discard changes in working directory)
modified: ../../day4/hello-world/app.yaml
bash-3.2$ git commit -m "fixed 500 error (issue #1)"
issue number
doing this automatically connects commit to issue in github
bash /Users/caleb
bash-3.2$ go get code.google.com/p/go-uuid/uuid
go: missing Mercurial command. See http://golang.org/s/gogetcmd
package code.google.com/p/go-uuid/uuid: exec: "hg": executable file not found in $PATH
bash-3.2$ go get github.com/nu7hatch/gouuid
bash-3.2$ go get github.com/gorilla/sessions
bash-3.2$

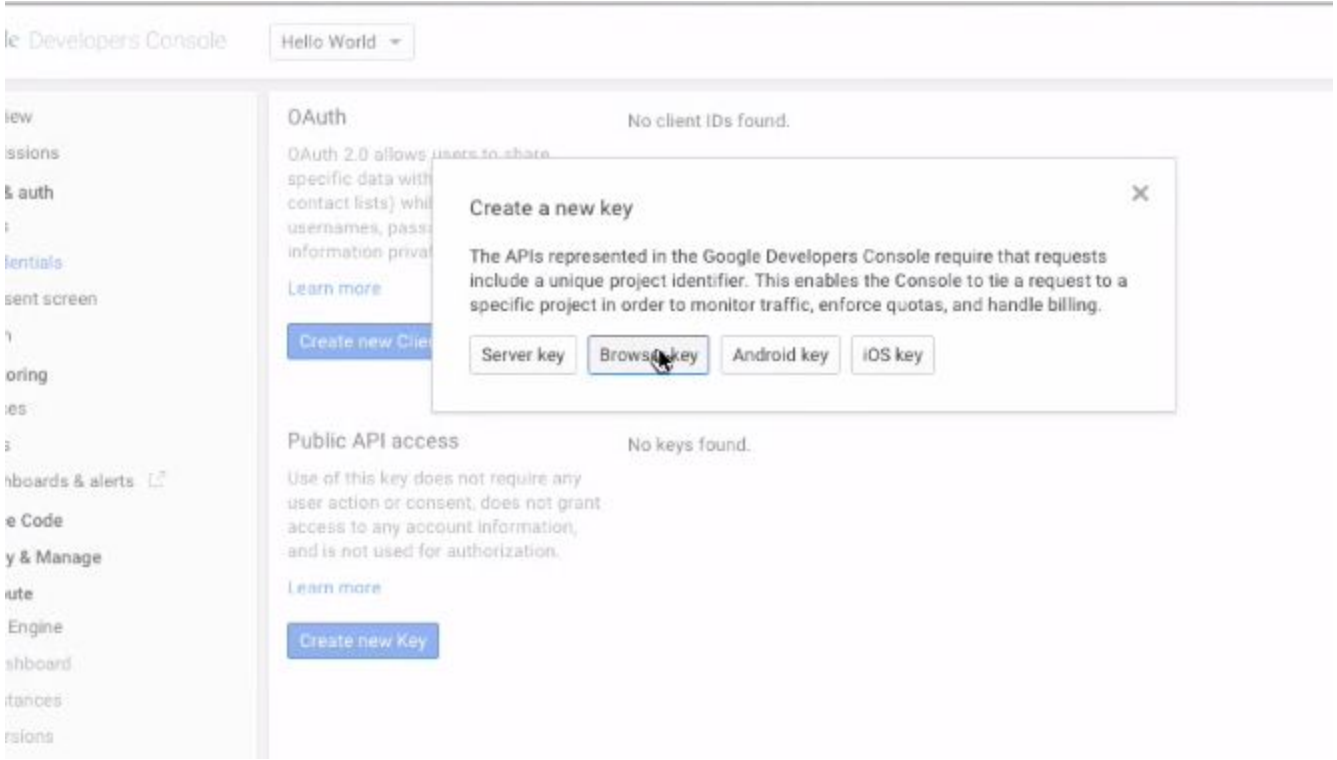
API’s


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APIs
Credentials
Push
Monitoring
Source Code
Cloud Launcher
Deployments
Compute
Networking
Storage
Big Data
API Library
Enabled APIs (6)
Search all 100+ APIs
Popular APIs
Google Cloud APIs
Compute Engine API
BigQuery API
Cloud Storage API
Cloud Datastore API
Cloud Deployment Manager API
Cloud DNS API
More
Google Maps APIs
Google Maps Android API
Google Maps SDK for iOS
Google Maps JavaScript API
Google Places API for Android
Google Places API for iOS
Google Maps Roads API
More
Google Apps APIs
Drive API
Calendar API
Gmail API
Google Apps Marketplace SDK
Admin SDK
Contacts API
CalDAV API
Mobile APIs
Cloud Messaging for Android
Google Play Game Services
Google Play Developer API
Google Places API for Android
Social APIs
Google+ API
Blogger API
Google+ Pages API
Google+ Domains API
YouTube APIs
YouTube Data API
YouTube Analytics API
Advertising APIs
AdSense Management API
DCM/DFA Reporting And Trafficking API
Ad Exchange Seller API
Ad Exchange Buyer API
DoubleClick Search API
Analytics API
DoubleClick Bid Manager API
Other popular APIs
Translate API
Custom Search API
URL Shortener API
PageSpeed Insights API
Fusion Tables API
Web Fonts Developer API

google maps API


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Consent screen
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Traces
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Google Maps Embed API
The Google Maps Embed API lets you embed Google Maps in your own web pages with a simple HTML fragment.
Learn more


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Traces
Logs
Dashboards & alerts
Source Code
Deploy & Manage
Compute
App Engine
Dashboard
OAuth
OAuth 2.0 allows users to share specific data with you (for example, contact lists) while keeping their usernames, passwords, and other information private.
Learn more
Create new Client ID
No client IDs found.
Public API access
Use of this key does not require any user action or consent, does not grant access to any account information, and is not used for authorization.
Learn more
Create new Key
No keys found.


le Developers Console
Hello World
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sent screen
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Engine
shboard
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rsions
OAuth
No client IDs found.
Learn more
Create new Clie
Create a new key
The APIs represented in the Google Developers Console require that requests include a unique project identifier. This enables the Console to tie a request to a specific project in order to monitor traffic, enforce quotas, and handle billing.
Server key
Browser key
Android key
iOS key
Public API access
No keys found.
Use of this key does not require any user action or consent, does not grant access to any account information, and is not used for authorization.
Learn more
Create new Key

Create a browser key and configure allowed referrers

This key can be deployed in your clients' applications.

API requests are sent directly to Google from your clients' browsers. Learn more

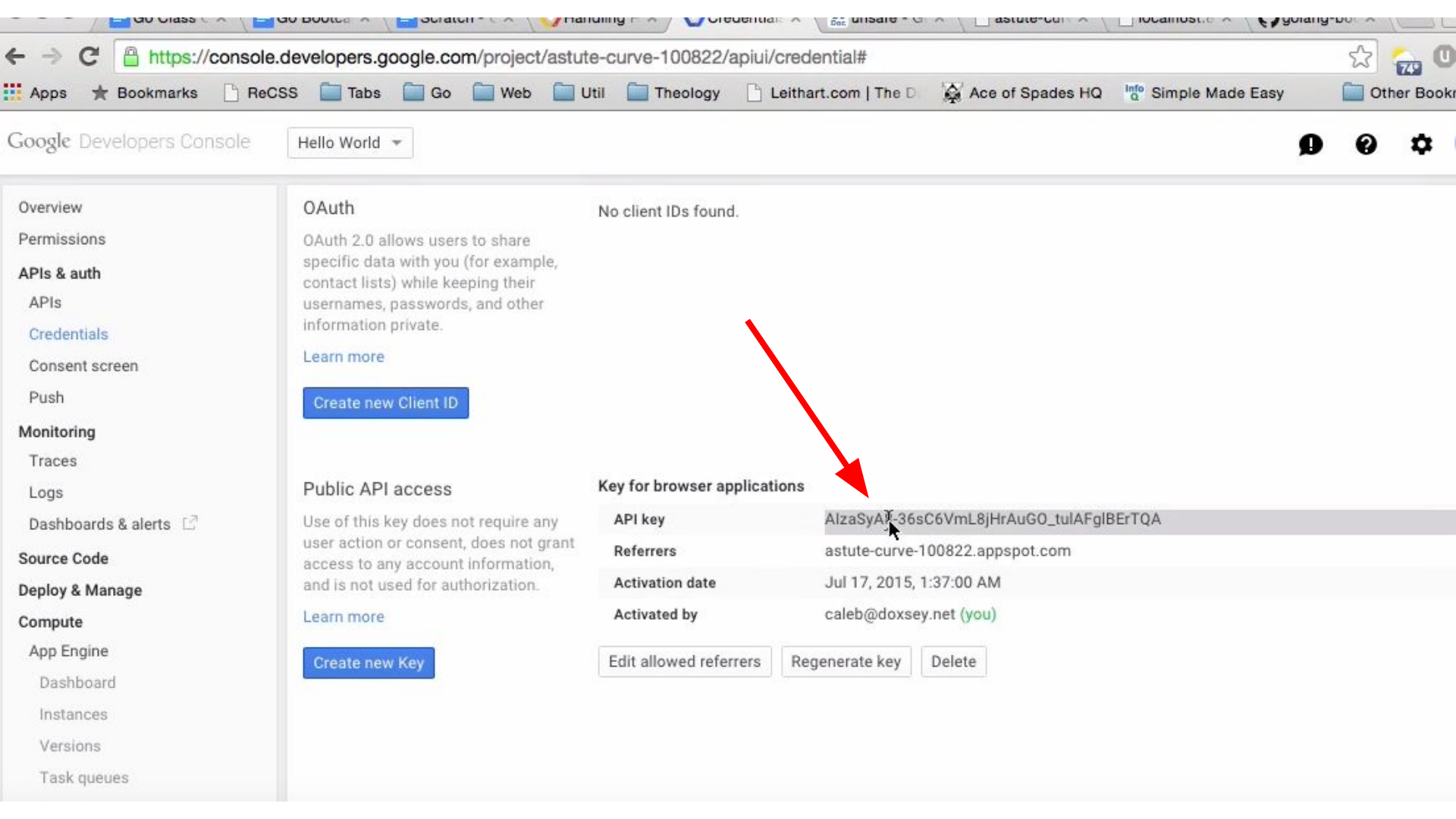
Accept requests from these HTTP referrers (web sites) (Optional)

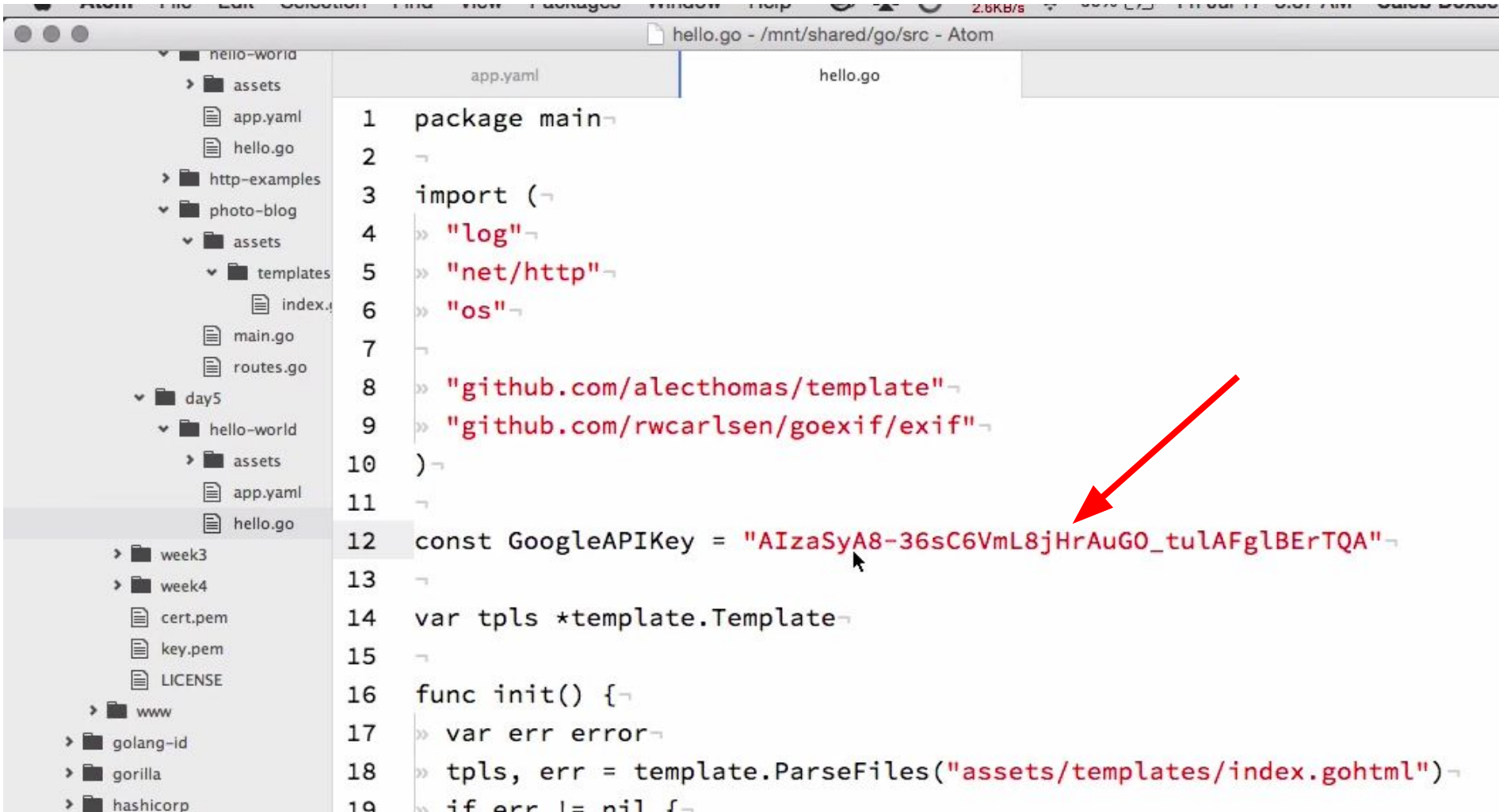
One URL or pattern per line. Example: *.example.com/*

Or if you leave this blank, requests will be accepted from any referer. Be sure to add referers before using this key in production.

astute-curve-100822.appspot.com

Create Cancel


https://console.developers.google.com/project/astute-curve-100822/apiui/credential#
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Ace of Spades HQ
Simple Made Easy
Other Bookm
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App Engine
Dashboard
Instances
Versions
Task queues
OAuth
OAuth 2.0 allows users to share specific data with you (for example, contact lists) while keeping their usernames, passwords, and other information private.
Learn more
Create new Client ID
No client IDs found.
Public API access
Use of this key does not require any user action or consent, does not grant access to any account information, and is not used for authorization.
Learn more
Create new Key
Key for browser applications
API key
AIzaSyA[redacted]-36sC6VmL8jHrAuGO_tulAFglBErTQA
Referrers
astute-curve-100822.appspot.com
Activation date
Jul 17, 2015, 1:37:00 AM
Activated by
caleb@doxsey.net (you)
Edit allowed referrers
Regenerate key
Delete


hello.go - /mnt/shared/go/src - Atom
hello-world
assets
app.yaml
hello.go
http-examples
photo-blog
assets
templates
index.
main.go
routes.go
day5
hello-world
assets
app.yaml
hello.go
week3
week4
cert.pem
key.pem
LICENSE
www
golang-id
gorilla
hashicorp
app.yaml
hello.go
1 package main
2
3 import (
4 "log"
5 "net/http"
6 "os"
7
8 "github.com/alecthomas/template"
9 "github.com/rwcarlsen/goexif/exif"
10)
11
12 const GoogleAPIKey = "AIzaSyA8-36sC6VmL8jHrAuGO_tulAFglBErTQA"
13
14 var tpls *template.Template
15
16 func init() {
17 var err error
18 tpls, err = template.ParseFiles("assets/templates/index.gohtml")

hello.go

```go
import (
    "log"
    "net/http"
    "os"
    "github.com/rwcarlsen/goexif/exif"
    "html/template"
)

const GoogleAPIKey = "AIzaSyDpMNCWNz2UENVGQOS6zMFvtLsXn0zMBf4"

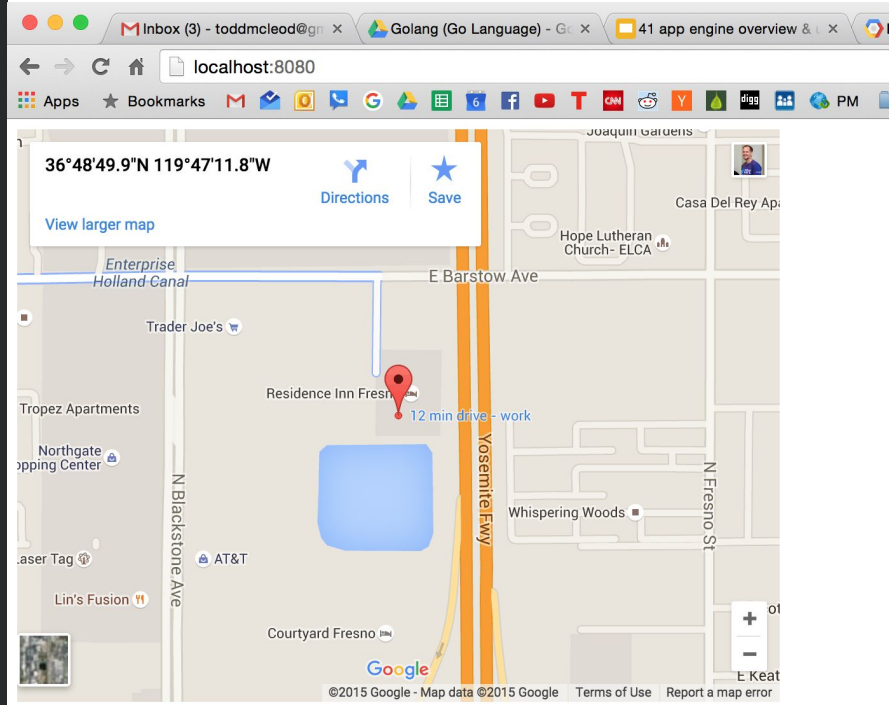
var tpls *template.Template

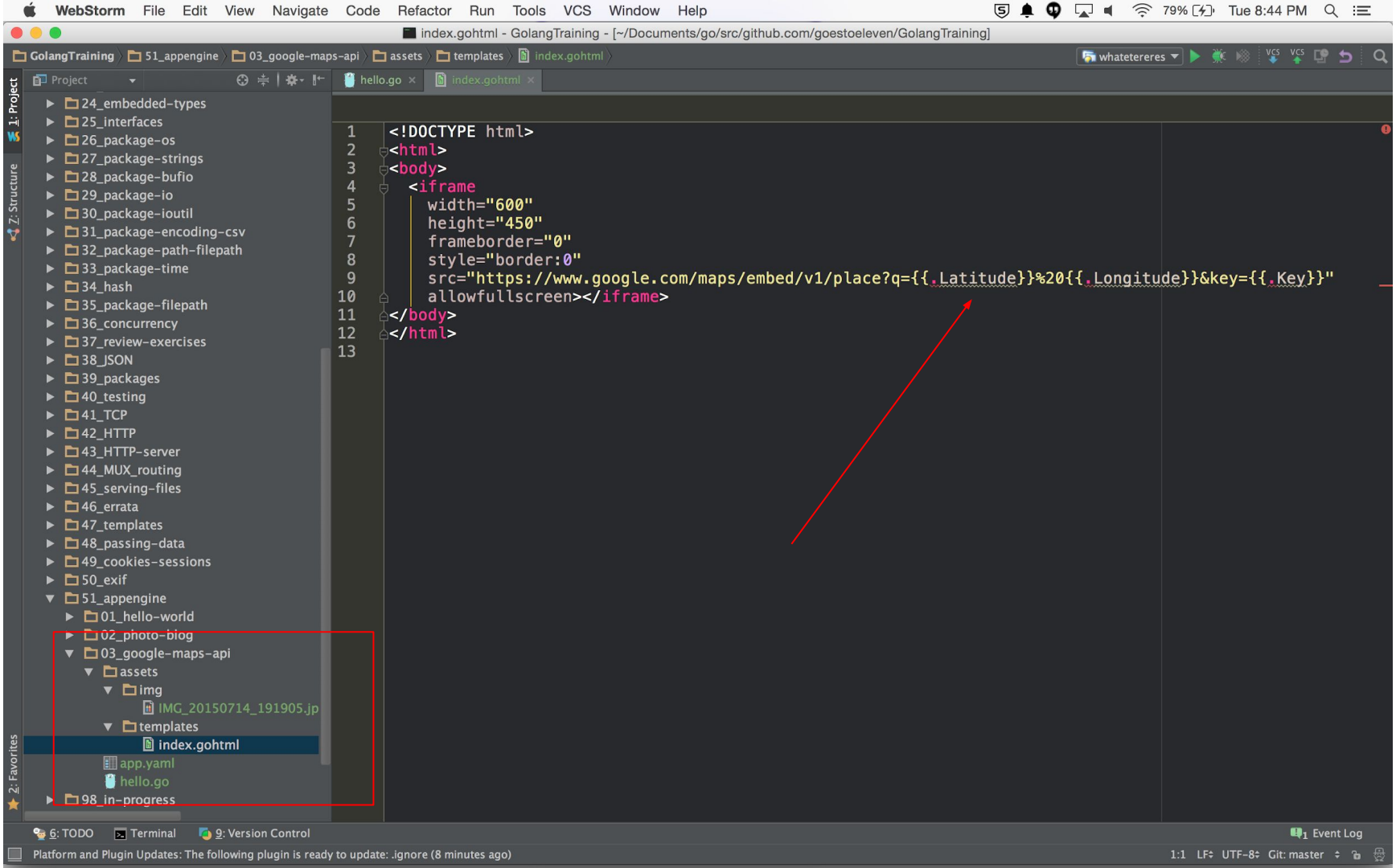
func init() {
    var err error
    tpls, err = template.ParseFiles("assets/templates/index.gohtml")
    if err != nil {
        log.Fatalln(err)
    }
    http.HandleFunc("/", handleIndex)
}

func handleIndex(res http.ResponseWriter, req *http.Request) {

    src, _ := os.Open("assets/img/IMG_20150714_191905.jpg")
    defer src.Close()
    x, _ := exif.Decode(src)
    lat, lon, _ := x.LatLong()

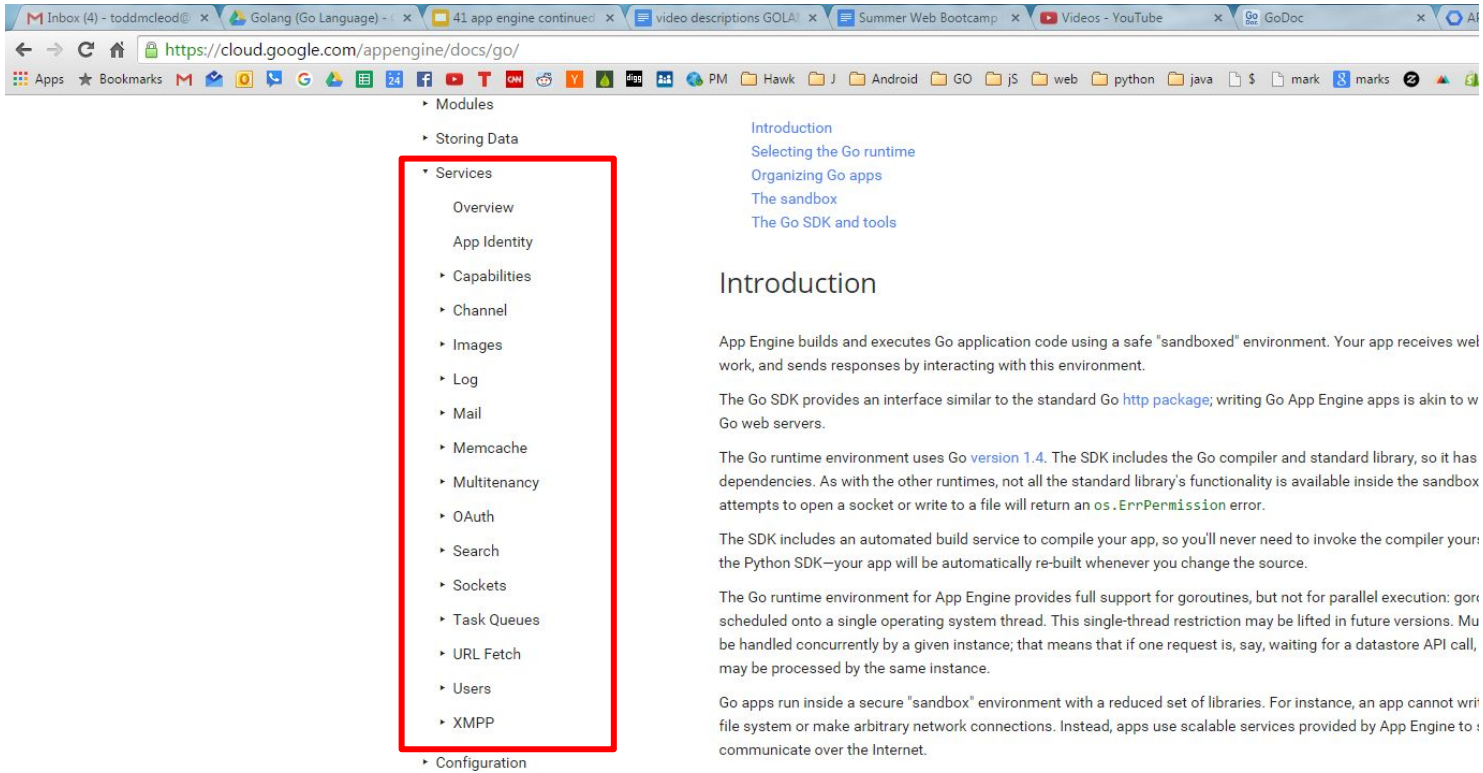
    var model struct {
        Latitude, Longitude float64
        Key                 string
    }
    model.Latitude = lat
    model.Longitude = lon
    model.Key = GoogleAPIKey
    err := tpls.ExecuteTemplate(res, "index.gohtml", model)
    if err != nil {
        http.Error(res, err.Error(), 500)
    }
}
```



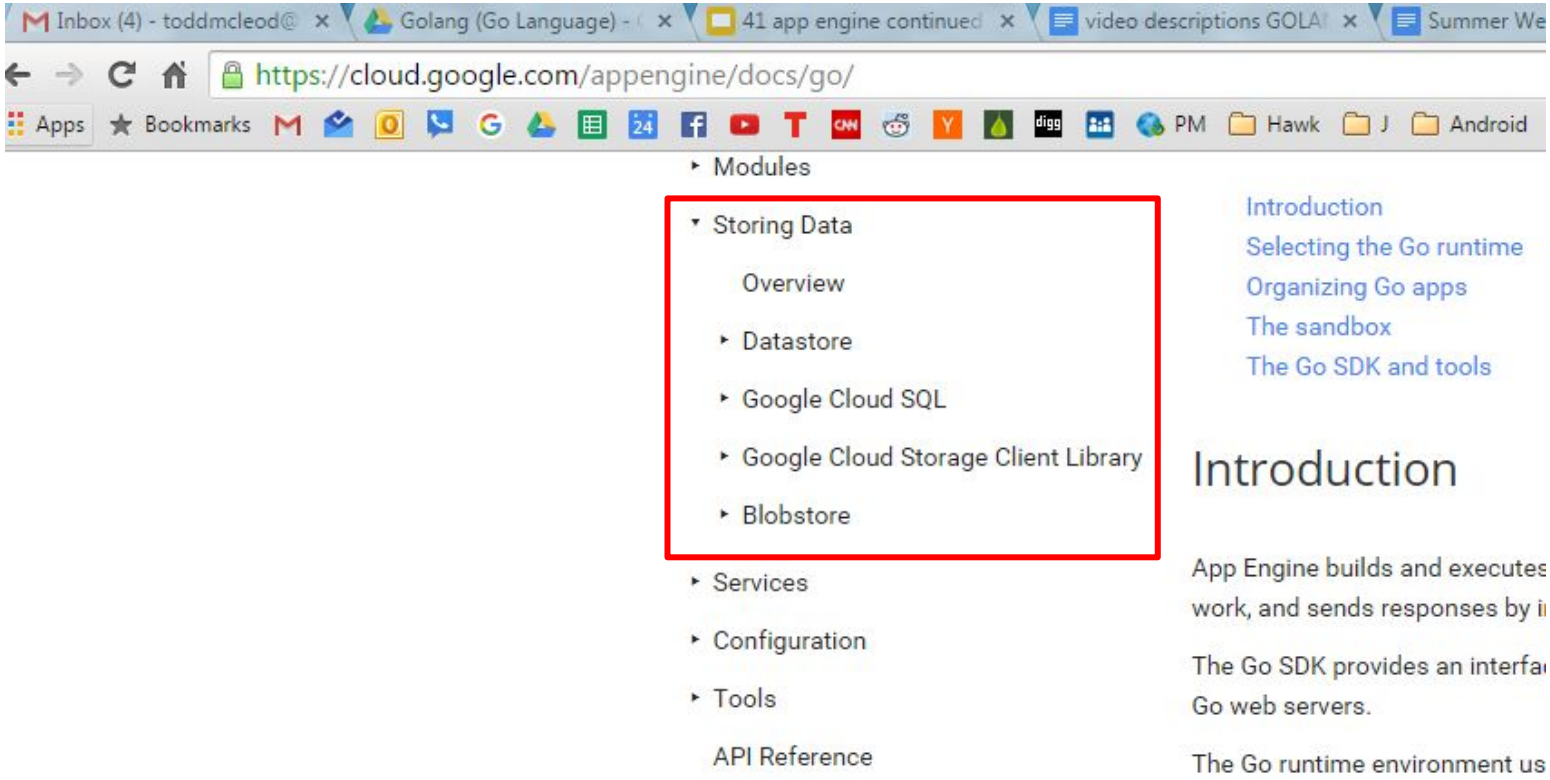

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79% Tue 8:44 PM
index.gohtml - GolangTraining - [~/Documents/go/src/github.com/goestoeleven/GolangTraining]
GolangTraining 51_appengine 03_google-maps-api assets templates index.gohtml
whatetereres
Project
hello.go
index.gohtml
24_embedded-types
25_interfaces
26_package-os
27_package-strings
28_package-bufio
29_package-io
30_package-ioutil
31_package-encoding-csv
32_package-path-filepath
33_package-time
34_hash
35_package-filepath
36_concurrency
37_review-exercises
38_JSON
39_packages
40_testing
41_TCP
42_HTTP
43_HTTP-server
44_MUX_routing
45_serving-files
46_errata
47_templates
48_passing-data
49_cookies-sessions
50_exif
51_appengine
01_hello-world
02_photo-biog
03_google-maps-api
assets
img
IMG_20150714_191905.jp
templates
index.gohtml
app.yaml
hello.go
98_in-progress
1 <!DOCTYPE html>
2 <html>
3 <body>
4 <iframe
5 width="600"
6 height="450"
7 frameborder="0"
8 style="border:0"
9 src="https://www.google.com/maps/embed/v1/place?q={{.Latitude}}%20{{.Longitude}}&key={{.Key}}"
10 allowfullscreen></iframe>
11 </body>
12 </html>
13
6: TODO
Terminal
9: Version Control
Event Log
Platform and Plugin Updates: The following plugin is ready to update: .ignore (8 minutes ago)
1:1 LF UTF-8 Git: master
1: Project
Z: Structure
2: Favorites

App Engine Services

App engine is made up of a lot of services



Data Storage



most of your app is

- go code
- html, css, javascript
- using the datastore
  - storing things in the datastore
  - retrieving things from the datastore

“Your go code is like the plumbing, connecting your html / css / javascript and the datastore”

USERS
app engine service

logs users in
using their google account

main.go

```go
package main

import (
    "fmt"
    "net/http"

    "google.golang.org/appengine"
    "google.golang.org/appengine/user"
)

func index(res http.ResponseWriter, req *http.Request) {
    ctx := appengine.NewContext(req)
    u := user.Current(ctx)
    url, _ := user.LogoutURL(ctx, "/")
    res.Header().Set("Content-Type", "text/html")
    fmt.Fprintf(res, `Welcome, %s! (<a href="%s">sign out</a>)`, u, url)

}

func admin(res http.ResponseWriter, req *http.Request) {
    ctx := appengine.NewContext(req)
    u := user.Current(ctx)
    url, _ := user.LogoutURL(ctx, "/")
    res.Header().Set("Content-Type", "text/html")
    fmt.Fprintf(res, `Welcome ADMIN, %s! (<a href="%s">sign out</a>)`, u, url)

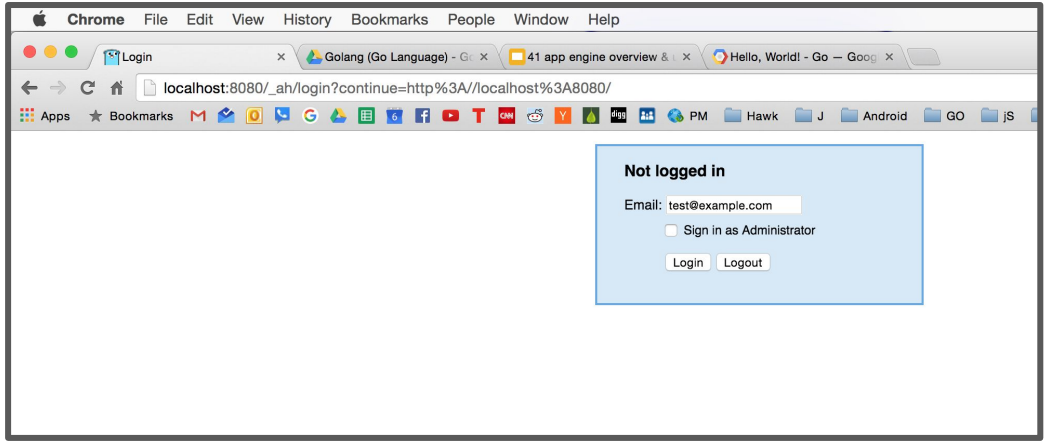
}

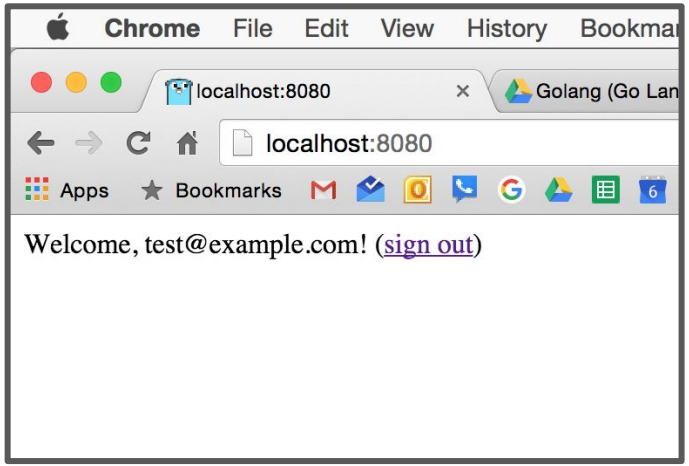
func init() {
    http.HandleFunc("/", index)
    http.HandleFunc("/admin/", admin)
}
```

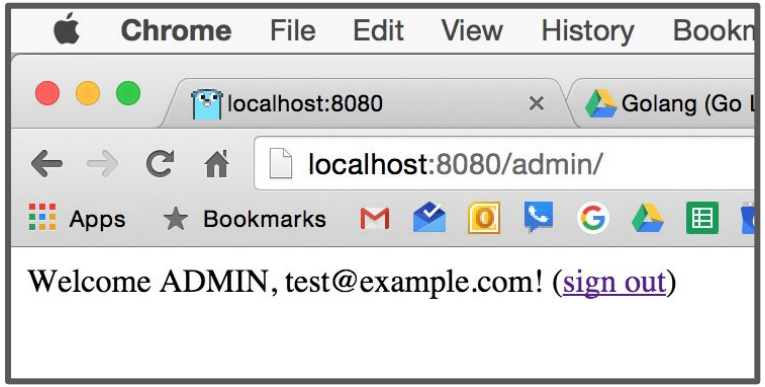
main.go | app.yaml

```yaml
application: astute-curve-100822
version: 1
runtime: go
api_version: go1

handlers:
- url: /admin/.*
  script: _go_app
  login: admin
- url: /.*
  script: _go_app
  login: required
```

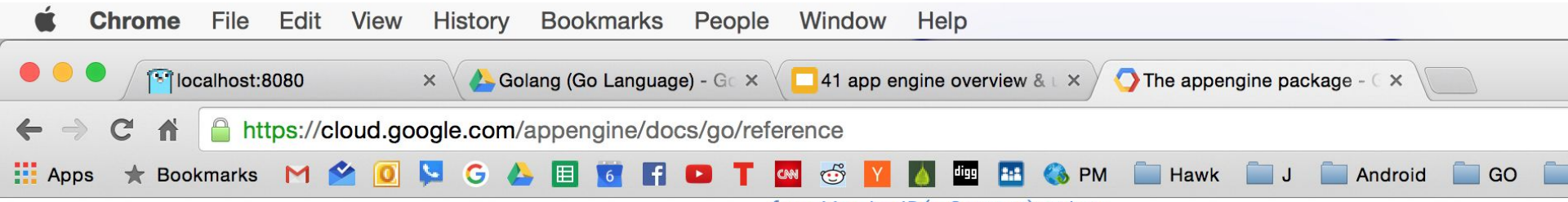

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Login
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Hello, World! - Go — Goog
localhost:8080/_ah/login?continue=http%3A//localhost%3A8080/
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Bookmarks
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Hawk
J
Android
GO
jS
Not logged in
Email: test@example.com
Sign in as Administrator
Login
Logout


Chrome File Edit View History Bookmar
localhost:8080
Golang (Go Lan
localhost:8080
Apps
Bookmarks
Welcome, test@example.com! (sign out)


Chrome File Edit View History Bookn
localhost:8080
Golang (Go
localhost:8080/admin/
Apps
Bookmarks
Welcome ADMIN, test@example.com! (sign out)

Context

- a piece of information that travels along with all of the API requests



func VersionID(c Context) string
type BlobKey
type Certificate
type Context
  func Namespace(c Context, namespace string) (Context, error)
  func NewContext(req *http.Request) Context
  func Timeout(c Context, d time.Duration) Context
type GeoPoint
  func (g GeoPoint) Valid() bool
type MultiError
  func (m MultiError) Error() string



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https://cloud.google.com/appengine/docs/go/reference

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- Handling Requests
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- Modules
- Storing Data
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- Configuration
- Tools
- API Reference
- Release Notes

The appengine package

```
import "appengine"
```

Introduction

Package appengine provides basic functionality for Google App Engine.

For more information on how to write Go apps for Google App Engine, see: https://cloud.google.com/appengine/docs/go/

Index

func AccessToken(c Context, scopes ...string) (token string, expiry time.Time, err error)
func AppID(c Context) string
func BackendHostname(c Context, name string, index int) string
func BackendInstance(c Context) (name string, index int)
func Datacenter() string
func DefaultVersionHostname(c Context) string
func InstanceID() string
func IsCapabilityDisabled(err error) bool
func IsDevAppServer() bool

type Context

```
type Context interface {
    // Debugf formats its arguments according to the format, analogous to fmt.Printf,
    // and records the text as a log message at Debug level.
    Debugf(format string, args ...interface{})

    // Infof is like Debugf, but at Info level.
    Infof(format string, args ...interface{})

    // Warningf is like Debugf, but at Warning level.
    Warningf(format string, args ...interface{})

    // Errorf is like Debugf, but at Error level.
    Errorf(format string, args ...interface{})

    // Criticalf is like Debugf, but at Critical level.
    Criticalf(format string, args ...interface{})

    // Internal use only.
    Call(service, method string, in, out appengine_internal.ProtoMessage, opts *appengine_internal
    // Internal use only. Use AppID instead.
    FullyQualifiedAppID() string
    // Internal use only.
    Request() interface{}
}
```

doesn't provide me with much insight into context

Context represents the context of an in-flight HTTP request.

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- Tools

Introduction

Package user provides a client for App Engine's user authentication service.

Index

func Current(c appengine.Context) *User
func CurrentOAuth(c appengine.Context, scope string) (*User, error)
func IsAdmin(c appengine.Context) bool
func LoginURL(c appengine.Context, dest string) (string, error)
func LoginURLFederated(c appengine.Context, dest, identity string) (string, error)
func LogoutURL(c appengine.Context, dest string) (string, error)
func OAuthConsumerKey(c appengine.Context) (string, error)
type User
  func (u *User) String() string

func Current

```
func Current(c appengine.Context) *User
```

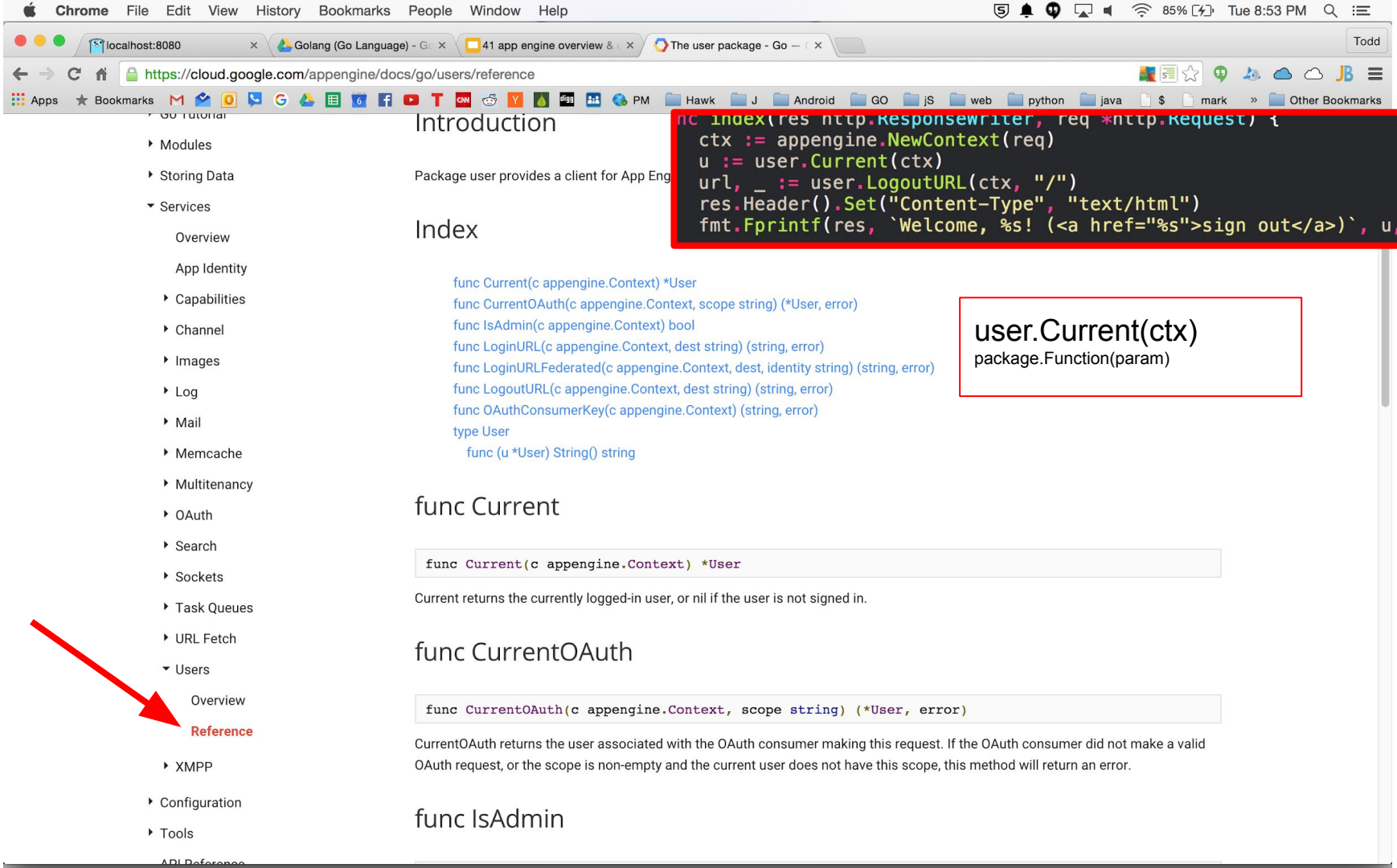
Current returns the currently logged-in user, or nil if the user is not signed in.

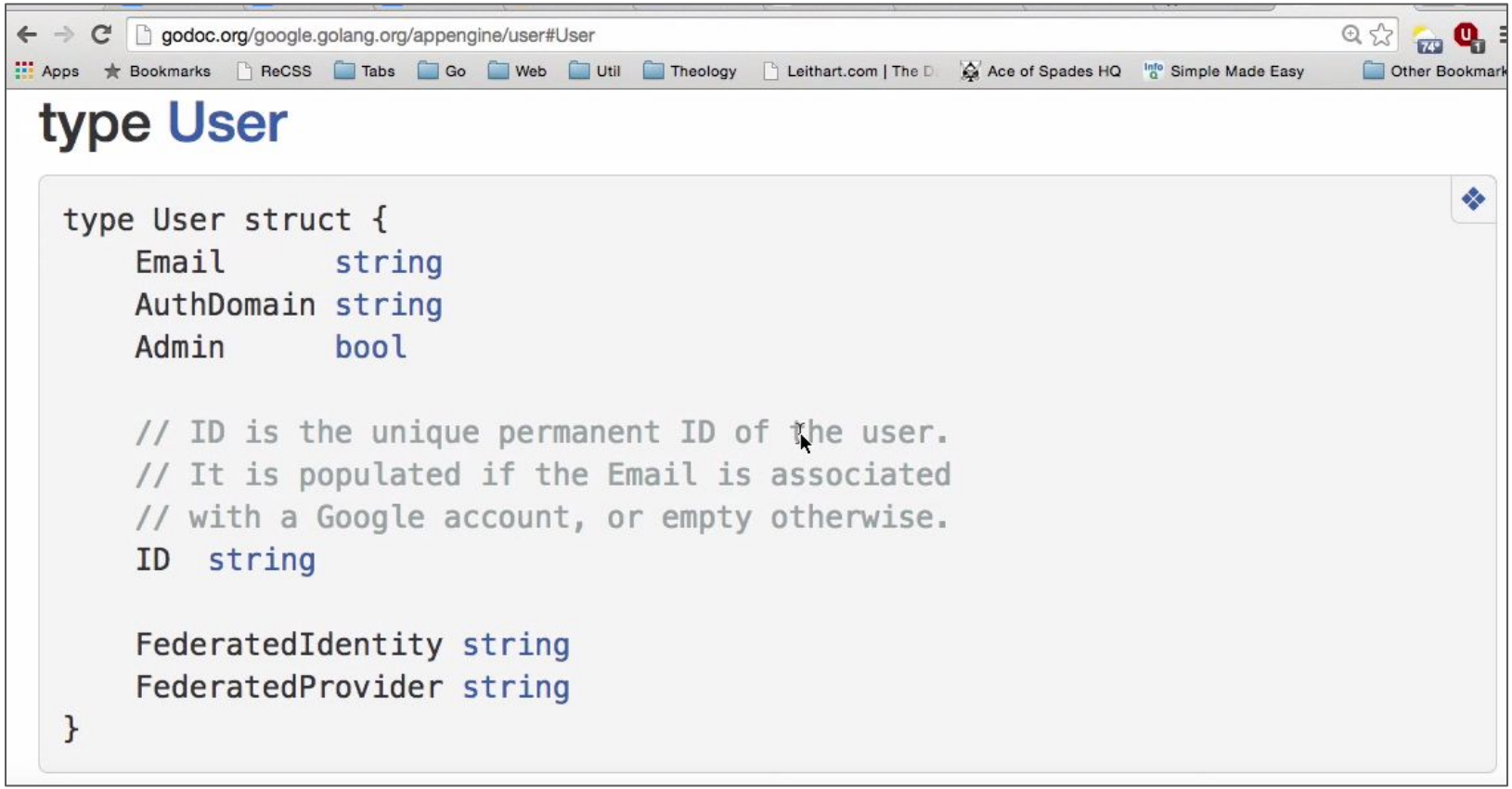
func CurrentOAuth

```
func CurrentOAuth(c appengine.Context, scope string) (*User, error)
```

CurrentOAuth returns the user associated with the OAuth consumer making this request. If the OAuth consumer did not make a valid OAuth request, or the scope is non-empty and the current user does not have this scope, this method will return an error.

func IsAdmin


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Introduction
Package user provides a client for App Eng
Index
func Current(c appengine.Context) *User
func CurrentOAuth(c appengine.Context, scope string) (*User, error)
func IsAdmin(c appengine.Context) bool
func LoginURL(c appengine.Context, dest string) (string, error)
func LoginURLFederated(c appengine.Context, dest, identity string) (string, error)
func LogoutURL(c appengine.Context, dest string) (string, error)
func OAuthConsumerKey(c appengine.Context) (string, error)
type User
func (u *User) String() string
func Current
func Current(c appengine.Context) *User
Current returns the currently logged-in user, or nil if the user is not signed in.
func CurrentOAuth
func CurrentOAuth(c appengine.Context, scope string) (*User, error)
CurrentOAuth returns the user associated with the OAuth consumer making this request. If the OAuth consumer did not make a valid OAuth request, or the scope is non-empty and the current user does not have this scope, this method will return an error.
func IsAdmin
ctx := appengine.NewContext(req)
u := user.Current(ctx)
url, _ := user.LogoutURL(ctx, "/")
res.Header().Set("Content-Type", "text/html")
fmt.Fprintf(res, `Welcome, %s! (<a href="%s">sign out</a>)`, u, url)
user.Current(ctx)
package.Function(param)


godoc.org/google.golang.org/appengine/user#User
Apps
Bookmarks
ReCSS
Tabs
Go
Web
Util
Theology
Leithart.com | The D
Ace of Spades HQ
Simple Made Easy
Other Bookmark
type User
type User struct {
Email string
AuthDomain string
Admin bool
// ID is the unique permanent ID of the user.
// It is populated if the Email is associated
// with a Google account, or empty otherwise.
ID string
FederatedIdentity string
FederatedProvider string
}

func (*User) String

```
func (u *User) String() string
```

String returns a displayable name for the user.

func LogoutURL

```
func LogoutURL(c appengine.Context, dest string) (string, error)
```

LogoutURL returns a URL that, when visited, signs the user out, then redirects the user to the URL specified by dest.

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Configuring with app.yaml

Python | Java | PHP | **Go**

A Go App Engine application can be configured by a file named `app.yaml` that specifies how URL paths correspond to request handlers and static files. It also contains information about the application code, such as the application ID and the latest version identifier.

> **Note:** If you created your project using the Google Developers Console, your project has a title and an ID. In the instructions that follow, the *project* title and ID can be used wherever an *application* title and ID are mentioned. They are the same thing.

About `app.yaml`
Required elements
Optional elements
Script handlers
Static file handlers
- Static directory handlers
- Static file pattern handlers
- Static cache expiration

Including handlers
Secure URLs
Requiring login or administrator status
Skipping files
Defining environment variables
Reserved URLs
Inbound services
Warmup requests
Custom error responses
Custom PageSpeed configuration (Deprecated)
Auto ID policy

About `app.yaml`

A Go app specifies runtime configuration, including versions and URLs, in a file named `app.yaml`. The following is an example of an

The following example requires a login for the `/profile/` directory and an administrator login for the `/admin/` directory:

```
handlers:

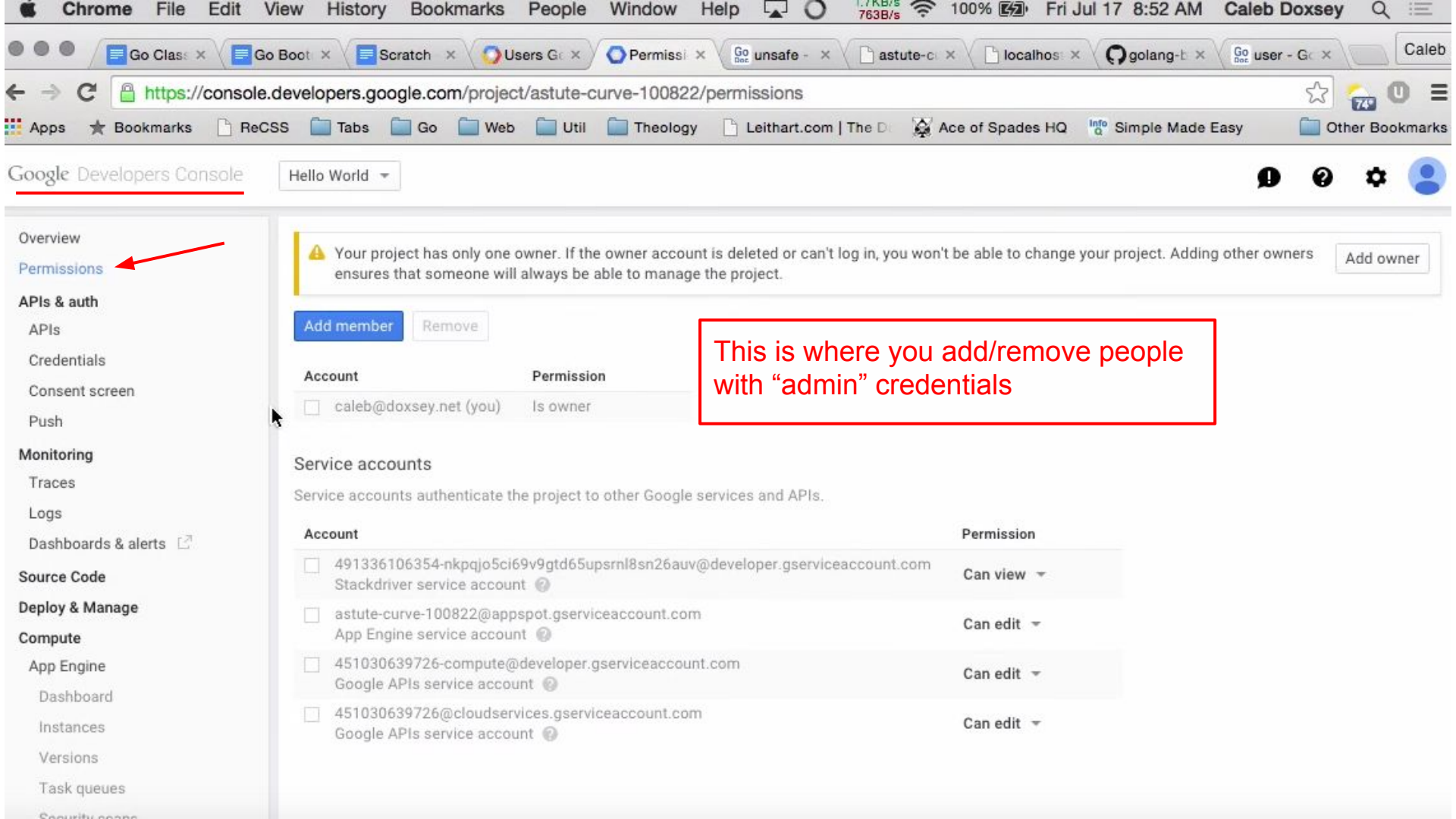
- url: /profile/.*
  script: _go_app
  login: required

- url: /admin/.*
  script: _go_app
  login: admin

- url: /.*
  script: _go_app
```

You can configure a handler to refuse access to protected URLs when the user is not signed in, instead of redirecting the user to the sign-in page, by adding `auth_fail_action: unauthorized` to the handler's configuration:

```
handlers:
- url: /secure_api/.*
  script: _go_app
  login: required
  auth_fail_action: unauthorized
```


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Fri Jul 17 8:52 AM Caleb Doxsey
https://console.developers.google.com/project/astute-curve-100822/permissions
Google Developers Console
Hello World
Overview
Permissions
APIs & auth
APIs
Credentials
Consent screen
Push
Monitoring
Traces
Logs
Dashboards & alerts
Source Code
Deploy & Manage
Compute
App Engine
Dashboard
Instances
Versions
Task queues
Your project has only one owner. If the owner account is deleted or can't log in, you won't be able to change your project. Adding other owners ensures that someone will always be able to manage the project.
Add owner
Add member
Remove
Account
Permission
caleb@doxsey.net (you)
Is owner
This is where you add/remove people with “admin” credentials
Service accounts
Service accounts authenticate the project to other Google services and APIs.
Account
Permission
491336106354-nkpqjo5ci69v9gtd65upsrnl8sn26auv@developer.gserviceaccount.com
Stackdriver service account
Can view
astute-curve-100822@appspot.gserviceaccount.com
App Engine service account
Can edit
451030639726-compute@developer.gserviceaccount.com
Google APIs service account
Can edit
451030639726@cloudservices.gserviceaccount.com
Google APIs service account
Can edit

What do you do
if you don’t want to use google logins?

roll your own session management

What do you do
if you don’t want to use google logins?